

Neural Bandit Algorithm Evaluation Framework

Graduate Research Project

AI & Quantum Computing Laboratory
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Research Framework Overview

This comprehensive evaluation framework provides rigorous analysis of neural bandit algorithms with clear categorical distinction between different operational environments:

- **Baseline Environment:** Optimal performance benchmark (Oracle)
- **Stochastic Environment:** Natural random failures and network noise
- **Adversarial Environment:** Strategic intelligent attacks and malicious targeting

Primary Research Questions

1. **Algorithm Robustness:** How do neural bandit algorithms perform across different threat models?
2. **Comparative Analysis:** Which algorithms demonstrate superior performance in specific scenarios?
3. **Quantified Performance:** What are the exact degradation metrics under adversarial conditions?
4. **Theoretical Validation:** Do experimental results align with established regret bounds?

Key Research Contributions

- **Systematic Environment Categorization:** Clear baseline/stochastic/adversarial taxonomy
- **Multi-Algorithm Comparative Testing:** Comprehensive evaluation across 6+ algorithms
- **Quantified Robustness Metrics:** Precise performance degradation measurements
- **Publication-Ready Analysis:** Academic-quality visualizations and statistical validation

Evaluation Methodology

The framework implements standardized testing protocols across three distinct categories:

- **Baseline:** Oracle performance establishing theoretical upper bounds
- **Stochastic:** Random environmental perturbations modeling realistic conditions
- **Adversarial:** Strategic attack scenarios simulating malicious interference

Each algorithm undergoes identical testing conditions enabling direct performance comparison and robustness quantification across all operational environments.

Environment Setup & Library Installation

```

In [1]: # @title Quantum MAB Models Evaluation Framework – Environment Setup
import os, sys
import warnings
warnings.filterwarnings('ignore')

# Get current directory
cur_dir = os.getcwd()
print(f"Current working directory: {cur_dir.split('/')[-1]}")

try:
    import google.colab          # Check if running in Colab
    from google.colab import drive
    drive.mount('/content/drive')

    # Change to framework directory (updated path)
    project_dir = '/content/drive/MyDrive/GA-Work/hybrid_variable_framework/Hybrid_MABs_'
    os.chdir(project_dir)
    print("Running in Google Colab")

    # Add source directory to path
    project_code_dir = os.path.join(project_dir, 'src')
    sys.path.append(project_code_dir)

except ImportError:
    print("Running locally (not in Colab)")
    # For local development, assume current directory structure
    pass

print(f"Now working from: {os.getcwd().split('/')[-1]}")

# Verify Python version and framework initialization
import sys
print(f"Python version: {sys.version.split()[0]}")
print("Quantum MAB Models Evaluation Framework – Environment Ready")

```

```

Current working directory: Hybrid_MABs_Evaluation_Framework
Running locally (not in Colab)
Now working from: Hybrid_MABs_Evaluation_Framework
Python version: 3.12.11
Quantum MAB Models Evaluation Framework – Environment Ready

```

```

In [2]: # @title Framework Dependencies Installation

# Install core scientific computing libraries for MAB evaluation
!pip install torch torchvision numpy matplotlib seaborn pandas tqdm scipy scikit-learn -

# Verify installations
import torch
import numpy as np
import matplotlib.pyplot as plt
from tqdm import tqdm
import seaborn as sns
import pandas as pd

print("Framework dependencies installed successfully")
print(f"PyTorch version: {torch.__version__}")
print(f"NumPy version: {np.__version__}")
print(f"Matplotlib version: {plt.matplotlib.__version__}")
print("Quantum MAB Models Evaluation Framework – Ready for model testing")

```

Threat Model Classification Framework

Systematic Environment Taxonomy

This research framework establishes precise categorical distinctions for quantum network evaluation environments, addressing previous ambiguity in threat model classification:

Environmental Categories

Environment	Implementation	Threat Characteristic	Research Application
Baseline	none	Deterministic optimal performance	Theoretical upper bound
Stochastic	stochastic / random	Natural random failures	Realistic network conditions
Adversarial	markov	Oblivious strategic attacks	Pattern-based targeting
Adversarial	adaptive	Responsive strategic attacks	Feedback-driven targeting
Adversarial	onlineadaptive	Real-time strategic attacks	Dynamic threat adaptation

Research Contribution

This framework addresses a critical gap in existing literature where random network failures were often conflated with intentional adversarial attacks. The systematic categorization enables:

- **Precise Robustness Quantification:** Exact performance degradation measurements across threat categories
- **Comparative Algorithm Analysis:** Direct performance comparison under identical threat conditions
- **Theoretical Validation:** Empirical verification of regret bounds across different adversarial models
- **Reproducible Research Standards:** Standardized evaluation protocols for quantum network algorithms

Methodological Significance

Previous research often lacked clear distinction between stochastic and adversarial environments, limiting the ability to assess true algorithm robustness under intentional attacks versus natural network degradation. This framework provides the necessary precision for rigorous academic evaluation of quantum routing algorithms.

In [3]:

```
# @title Quantum MAB Models Evaluation Framework – Core Components Import

import importlib
import os
import numpy as np
import torch
```

```

# Install pmdarima for time series analysis (used in some models)
try:
    import pmdarima as pm
except ImportError:
    !pip install pmdarima -q
    import pmdarima as pm

print(f"Now working from: {os.getcwd().split('/')[ -1 ]}")

# Import and reload framework components
try:
    # Core algorithm implementations
    import quantum_algorithms
    importlib.reload(quantum_algorithms)
    import quantum_environment
    importlib.reload(quantum_environment)

    # Framework evaluation and testing components
    import quantum_framework_updated
    importlib.reload(quantum_framework_updated)
    import quantum_experiments_updated
    importlib.reload(quantum_experiments_updated)

    import quantum_evaluator_visualizer
    importlib.reload(quantum_evaluator_visualizer)
    # # Stochastic testing module (renamed from test_stochastic_vs_adversarial)
    import quantum_models_stochastic_evaluation
    importlib.reload(quantum_models_stochastic_evaluation)

    print("Quantum MAB Models Evaluation Framework – All components imported successfully")

    # Try to verify framework configuration (graceful handling if missing)
    try:
        # Try to import from the updated stochastic evaluation module first
        from quantum_models_stochastic_evaluation import AVAILABLE_MODELS, FRAMEWORK_CONFIG
        print(f"Framework configured for: {FRAMEWORK_CONFIG.get('primary_environment', 'default')}")

        total_models = sum(len(models) for models in AVAILABLE_MODELS.values())
        print(f"Available models for testing: {total_models}")

    except ImportError:
        # Fallback: try from quantum_algorithms
        try:
            from quantum_algorithms import AVAILABLE_MODELS, FRAMEWORK_CONFIG
            print(f"Framework configured for: {FRAMEWORK_CONFIG.get('primary_environment', 'default')}")

            total_models = sum(len(models) for models in AVAILABLE_MODELS.values())
            print(f"Available models for testing: {total_models}")

        except ImportError:
            print("\t Framework configuration (AVAILABLE_MODELS, FRAMEWORK_CONFIG) not available")
            print("\t This is expected during development – proceeding without configuration")

except ImportError as e:
    print(f"❌ Import error encountered: {e}")
    print(f"⚠️ Note: Some modules may need to be created or paths adjusted.")
    print(f"🔄 Proceeding with fallback component loading...")

# Fallback: Try importing individual components
try:
    import quantum_algorithms
    import quantum_evaluator_visualizer

```

```

print("✅ Core components loaded successfully with fallback method")

# Try to get basic info if available
try:
    # Check if we can import classes from visualizer
    from quantum_evaluator_visualizer import QuantumEvaluatorVisualizer
    print("📊 Visualizer class available: EnhancedQuantumVisualizer")
except ImportError:
    print("⚠️ Visualizer class not available – may need updating")

except ImportError as fallback_error:
    print(f"❌ Fallback import also failed: {fallback_error}")
    print("🔧 Manual module creation may be required")

# Display current module status
print("\n" + "="*50)
print("MODULE STATUS SUMMARY:")
print("="*50)

modules_to_check = [
    'quantum_algorithms',
    'quantum_environment',
    'quantum_framework_updated',
    'quantum_experiments_updated',
    'quantum_evaluator_visualizer',
    'quantum_models_stochastic_evaluation'
]

for module_name in modules_to_check:
    try:
        module = __import__(module_name)
        print(f"✅ {module_name:<30} – Loaded")
    except ImportError:
        print(f"❌ {module_name:<30} – Not available")

print("="*50)

```

Now working from: Hybrid_MABs_Evaluation_Framework

PyTorch version: 2.8.0

NumPy version: 1.26.4

Using device: cpu

PyTorch version: 2.8.0

NumPy version: 1.26.4

Using device: cpu

Quantum MAB Models Evaluation Framework – All components imported successfully

Framework configured for: stochastic evaluation

Available models for testing: 11

```

=====
MODULE STATUS SUMMARY:
=====
✅ quantum_algorithms           – Loaded
✅ quantum_environment          – Loaded
✅ quantum_framework_updated    – Loaded
✅ quantum_experiments_updated  – Loaded
✅ quantum_evaluator_visualizer – Loaded
✅ quantum_models_stochastic_evaluation – Loaded
=====

```

Theoretical Foundation and Algorithm Architecture

Algorithm Composition

The **EXPNeuralUCB** algorithm integrates two complementary theoretical frameworks:

- **EXP3 Component:** Exponential-weight group selection mechanism for adversarial robustness
- **NeuralUCB Component:** Neural network-based arm selection with rigorous uncertainty quantification

Theoretical Performance Guarantees

The algorithm maintains provable regret bounds under adversarial conditions:

$$[\mathbb{E}[\text{Regret}(T)] = O(\sqrt{KT\log T} + \sqrt{ST\log T})]$$

where:

- (K): Number of available arms (quantum routing paths)
- (S): Number of strategic groups (allocation strategies)
- (T): Time horizon (evaluation period)

Research Hypotheses

This evaluation framework tests three primary hypotheses regarding algorithm performance and robustness:

H1: Adversarial Superiority EXPNeuralUCB demonstrates measurably superior performance in adversarial environments compared to baseline neural bandit algorithms lacking adversarial robustness mechanisms.

H2: Bounded Performance Degradation

Performance degradation under adversarial attacks remains within acceptable bounds (< 15% relative to stochastic performance), validating the algorithm's practical applicability.

H3: Consistent Ranking Stability Algorithm performance rankings remain stable across varying adversarial attack intensities, indicating reliable robustness characteristics.

Experimental Validation Framework

The theoretical predictions are empirically validated through systematic evaluation across the defined environmental categories, providing quantitative evidence for the algorithm's adversarial robustness properties while maintaining performance guarantees under standard stochastic conditions.

```
In [4]: # @title Quantum MAB Models Evaluation Framework – Experimental Configuration

import quantum_experiments_updated
importlib.reload(quantum_experiments_updated)

from quantum_experiments_updated import NEURAL_MODELS, CONTEXTUAL_MODELS, INFORMED_CONTE
# Define models for comprehensive evaluation (expandable)
models = NEURAL_MODELS

# Framework-specific experimental configuration
FRAMEWORK_CONFIG = {
```

```

# Testing Configuration (lightweight for development)
'test_mode': True, # Set to False for final comprehensive evaluation
'base_frames': 3000, # Quick testing (increase to 4000+ for final runs)
'exp_num': 3, # Fast iteration (increase to 10+ for statistical significance)
'frame_step': 2000,
'models': models,

# Production Configuration (for final evaluation)
'prod_frames': 4000, # Full evaluation frames
'prod_experiments': 10, # Statistical significance
'frame_steps': [], # Multi-horizon testing

# Primary evaluation focus
'main_env': 'stochastic',
'eval_mod': 'comprehensive',
'main_model': 'CEXPNeuralUCB',

# Algorithm parameters (EXPNeuralUCB paper-compliant)
# Algorithm parameters (EXPNeuralUCB paper-compliant)
'alg_attrs': {
    'lambda_reg': 1.0, # Regularization parameter
    'gamma': 0.1, # EXP3 exploration rate
    'network_width': 128, # Neural network width
    'network_depth': 2, # Neural network depth
    'gradient_steps': 8, # Gradient descent steps
    'learning_rate': 1e-4 # Learning rate
},

# Environment parameters
'env_attrs': {
    'intensity': 0.25, # Natural failure rate for stochastic
    'base_seed': 12345,
    'reproducible': True
},

# Test scenarios for different evaluation modes
'scenarios': {
    'exp_focus': ['stochastic'], # Primary focus
    'stochastic_vs_baseline': ['none', 'stochastic'], # Comparison
    'comprehensive': ['none', 'stochastic', 'markov', 'adaptive'], # Full evaluation
    'adversarial': ['markov', 'adaptive', 'onlineadaptive'] # Future use
}
}

for exp_id in range(1, FRAMEWORK_CONFIG['exp_num']):
    FRAMEWORK_CONFIG['frame_steps'].append(FRAMEWORK_CONFIG['base_frames']+(FRAMEWORK_CO

# Dynamic configuration based on testing mode
current_frames = FRAMEWORK_CONFIG['base_frames'] if FRAMEWORK_CONFIG['test_mode'] else F
current_experiments = FRAMEWORK_CONFIG['exp_num'] if FRAMEWORK_CONFIG['test_mode'] else

# Display current configuration
print("QUANTUM MAB MODELS EVALUATION FRAMEWORK – CONFIGURATION")
print("=" * 70)
print(f"Models to evaluate: {len(models)} total")
print(f"Primary environment: {FRAMEWORK_CONFIG['main_env'].upper()}")

print(f"\nCURRENT SETTINGS ({'TESTING' if FRAMEWORK_CONFIG['test_mode'] else 'PRODUCTION'})")
print(f" • Frames per run: {current_frames}")
print(f" • Experiments per model: {current_experiments}")
print(f" • Expected runtime: {'~2–3 minutes' if FRAMEWORK_CONFIG['test_mode'] else '~30"}

```

```

print(f"\nALGORITHM PARAMETERS:")
for key, value in FRAMEWORK_CONFIG['alg_attrs'].items():
    print(f"    • {key}: {value}")

print(f"\nENVIRONMENT PARAMETERS:")
for key, value in FRAMEWORK_CONFIG['env_attrs'].items():
    print(f"    • {key}: {value}")

print(f"\nAVAILABLE TEST SCENARIOS:")
for scenario_name, environments in FRAMEWORK_CONFIG['scenarios'].items():
    print(f"    • {scenario_name}: {environments}")

print("\n" + "=" * 70)
print("DEVELOPMENT STRATEGY:")
if FRAMEWORK_CONFIG['test_mode']:
    print("TESTING MODE – Fast iteration for development and debugging")
    print("Switch test_mode to False for final comprehensive evaluation")
else:
    print("PRODUCTION MODE – Full statistical evaluation")
print("=" * 70)

print("\nConfiguration loaded successfully – Ready for model evaluation")

```

QUANTUM MAB MODELS EVALUATION FRAMEWORK – CONFIGURATION

=====

Models to evaluate: 3 total

Primary environment: STOCHASTIC

CURRENT SETTINGS (TESTING MODE):

- Frames per run: 3000
- Experiments per model: 3
- Expected runtime: ~2–3 minutes

ALGORITHM PARAMETERS:

- lambda_reg: 1.0
- gamma: 0.1
- network_width: 128
- network_depth: 2
- gradient_steps: 8
- learning_rate: 0.0001

ENVIRONMENT PARAMETERS:

- intensity: 0.25
- base_seed: 12345
- reproducible: True

AVAILABLE TEST SCENARIOS:

- exp_focus: ['stochastic']
- stochastic_vs_baseline: ['none', 'stochastic']
- comprehensive: ['none', 'stochastic', 'markov', 'adaptive']
- adversarial: ['markov', 'adaptive', 'onlineadaptive']

=====

DEVELOPMENT STRATEGY:

TESTING MODE – Fast iteration for development and debugging

Switch test_mode to False for final comprehensive evaluation

=====

Configuration loaded successfully – Ready for model evaluation

Comparative Analysis Framework: Stochastic versus Adversarial Environments

Research Focus

This evaluation constitutes the primary empirical contribution of the research: systematic quantification of algorithm performance across fundamentally different operational conditions that distinguish between natural system failures and intentional strategic attacks.

Environmental Characterization

Stochastic Environment

- **Operational Model:** Natural random failures representing realistic network degradation patterns
- **Attack Distribution:** Probabilistic failures following uniform random distribution
- **Research Significance:** Establishes baseline performance metrics under standard operational conditions

Adversarial Environment

- **Operational Model:** Strategic intelligent attacks systematically targeting algorithmic decision-making processes
- **Attack Distribution:** Adaptive targeting mechanisms that dynamically respond to observed algorithm behavior
- **Research Significance:** Evaluates robustness under worst-case strategic threat scenarios

Experimental Predictions

Based on the theoretical analysis and algorithm architecture, the following empirical outcomes are anticipated:

Performance Superiority Hypothesis EXPNeuralUCB will demonstrate measurably superior performance retention in adversarial environments relative to baseline neural bandit algorithms lacking specialized adversarial robustness mechanisms.

Bounded Degradation Hypothesis Performance degradation under adversarial conditions will remain within acceptable operational limits, specifically maintaining performance within 85% of stochastic environment baselines.

Stability Hypothesis Algorithm performance rankings will exhibit stability across varying adversarial attack intensities, indicating consistent robustness characteristics rather than scenario-dependent performance fluctuations.

Research Methodology

The comparative analysis employs identical experimental conditions across both environments, enabling precise quantification of performance degradation attributable to adversarial targeting while controlling for environmental variables and maintaining statistical rigor in the evaluation process.

In [5]: # @title Quantum MAB Models Stochastic Evaluation – Primary Framework Execution

```
# Reload modules to ensure latest versions
import quantum_evaluator_visualizer
importlib.reload(quantum_evaluator_visualizer)

# Import framework components using the correct class name
from quantum_evaluator_visualizer import QuantumEvaluatorVisualizer

print("=" * 70)
print("QUANTUM MAB MODELS EVALUATION FRAMEWORK – STOCHASTIC FOCUS")
print("=" * 70)

# Initialize framework evaluator
custom_model = "CEXPNeuralUCB"
viz = QuantumEvaluatorVisualizer()

print("Framework Configuration:")
print(f"\tPrimary Environment: {FRAMEWORK_CONFIG['main_env'].upper()}")
print(f"\tEvaluation Mode: {FRAMEWORK_CONFIG['eval_mod'].upper()}")
print(f"\tModels to Test: {len(models)}")
print("=" * 70)

# Define evaluation scenarios based on framework focus
if FRAMEWORK_CONFIG['main_env'] == 'stochastic':
    # Primary stochastic evaluation with optional comparison
    test_scenarios = {
        'stochastic': 'Stochastic (Natural Network Failures)',
        # 'none': 'Baseline (Optimal Conditions)' # For comparison
    }
    evaluation_type = "STOCHASTIC-FOCUSED"
else:
    # Fallback to stochastic vs adversarial for comparison
    test_scenarios = {
        'stochastic': 'Stochastic (Natural Network Failures)',
        'adaptive': 'Adversarial (Strategic Attacks)'
    }
    evaluation_type = "COMPARATIVE"

print(f"Executing {evaluation_type} EVALUATION:")
for scenario, description in test_scenarios.items():
    print(f"\t{scenario.upper()}: {description}")
print("=" * 70)

# Execute framework evaluation
try:
    # Use the correct method name that exists in the visualizer
    evaluator, comparison_results = viz.run_stochastic_test(
        base_frames=FRAMEWORK_CONFIG['base_frames'],
        frame_step=FRAMEWORK_CONFIG['frame_step'],
        runs=FRAMEWORK_CONFIG['exp_num'],
        test_scenarios=test_scenarios,
        selected_model=custom_model, # Changed from selected_model
        models=models
    )

    # Enhanced results summary
    print("\nFRAMEWORK EVALUATION RESULTS:")
    print("-" * 50)

    if comparison_results:
```

```

for scenario in test_scenarios.keys():
    if scenario in comparison_results:
        all_experiments = comparison_results[scenario]
        num_experiments = len(all_experiments)

        # Track each winner
        winner_stats = {}
        oracle_avg_reward = 0
        for exp_id, exp_data in all_experiments.items():
            winner = exp_data['winner']
            oracle_avg_reward += exp_data['oracle_reward']
            winner_efficiency = exp_data['winner_efficiency'] # wherever it's store
            winner_gap = exp_data['results'][winner]['gap'] # wherever it's sto
            winner_reward = exp_data['results'][winner]['final_reward'] # where

            if winner not in winner_stats:
                winner_stats[winner] = {'count': 0, 'avg_efficiency': 0, 'avg_re

            winner_stats[winner]['count'] += 1
            winner_stats[winner]['avg_gap'] += winner_gap
            winner_stats[winner]['avg_reward'] += winner_reward
            winner_stats[winner]['avg_efficiency'] += winner_efficiency

        # Find most common winner
        overall_winner = max(winner_stats, key=lambda x: winner_stats[x]['count']

        # Calculate averages using num_experiments
        oracle_avg_reward = oracle_avg_reward / num_experiments
        avg_gap = winner_stats[overall_winner]['avg_gap'] / num_experiments
        avg_reward = winner_stats[overall_winner]['avg_reward'] / num_experiment
        avg_efficiency = winner_stats[overall_winner]['avg_efficiency'] / num_ex

        print(f"\tRecommended model for stochastic environments: {overall_winner}")
        print(f"\tFramework validated stochastic performance")
        print(f"\tOracle Baseline: {oracle_avg_reward:.3f}")
        print(f"\tFinal Reward: {avg_reward:.3f}")
        print(f"\tWinner Avg Gap: {avg_gap:.1f}%")
        print(f"\tWinner Avg Efficiency: {avg_efficiency:.1f}%")

# Framework-specific insights
if FRAMEWORK_CONFIG['main_env'] == 'stochastic':
    print(f"\nSTOCHASTIC EVALUATION INSIGHTS:")
    scenario = 'stochastic'
    if scenario in comparison_results:
        all_experiments = comparison_results[scenario]
        num_experiments = len(all_experiments)

        # Track each winner
        winner_stats = {}
        oracle_avg_reward = 0
        for exp_id, exp_data in all_experiments.items():
            winner = exp_data['winner']
            oracle_avg_reward += exp_data['oracle_reward']
            winner_efficiency = exp_data['winner_efficiency'] # wherever it's store
            winner_gap = exp_data['results'][winner]['gap'] # wherever it's sto
            winner_reward = exp_data['results'][winner]['final_reward'] # where

            if winner not in winner_stats:
                winner_stats[winner] = {'count': 0, 'avg_efficiency': 0, 'avg_re

            winner_stats[winner]['count'] += 1
            winner_stats[winner]['avg_gap'] += winner_gap

```

```

        winner_stats[winner]['avg_reward'] += winner_reward
        winner_stats[winner]['avg_efficiency'] += winner_efficiency

    # Find most common winner
    overall_winner = max(winner_stats, key=lambda x: winner_stats[x]['count'])

    # Calculate averages using num_experiments
    oracle_avg_reward = oracle_avg_reward / num_experiments
    avg_gap = winner_stats[overall_winner]['avg_gap'] / num_experiments
    avg_reward = winner_stats[overall_winner]['avg_reward'] / num_experiments
    avg_efficiency = winner_stats[overall_winner]['avg_efficiency'] / num_experiments

    print(f"\tRecommended model for stochastic environments: {overall_winner}")
    print(f"\tFramework validated stochastic performance")
    print(f"\tOracle Baseline: {oracle_avg_reward:.3f}")
    print(f"\tFinal Reward: {avg_reward:.3f}")
    print(f"\tWinner Avg Gap: {avg_gap:.1f}%")
    print(f"\tWinner Avg Efficiency: {avg_efficiency:.1f}%")

print(f"\nQuantum MAB Models Evaluation Framework – {evaluation_type} evaluation complete")
print("Generated visualizations saved for analysis")

except Exception as e:
    print(f"Evaluation error: {e}")
    print("This may indicate missing framework components or configuration issues")

```

QUANTUM MAB MODELS EVALUATION FRAMEWORK - STOCHASTIC FOCUS

Framework Configuration:

Primary Environment: STOCHASTIC
Evaluation Mode: COMPREHENSIVE
Models to Test: 3

Executing STOCHASTIC-FOCUSED EVALUATION:

STOCHASTIC: Stochastic (Natural Network Failures)

Running Quantum MAB Models Evaluation...

Multi-Run Evaluator Initialized

Environment Type: stochastic

Frame Range: 3000 -> 7000 (step: 2000)

COMPREHENSIVE MODEL EVALUATION

Testing algorithm performance against:

- Stochastic: Natural quantum decoherence and network failures
- Baseline: Optimal conditions for validation

TESTING ENVIRONMENT SCENARIO: Stochastic (Natural Network Failures)

STARTING EXPERIMENTS: STOCHASTIC

Category: Stochastic (Natural Random Failures)

EXPERIMENT 1: 3000 frames

EXPERIMENT: 3000 frames

Attack Type: stochastic

Category: Stochastic

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	3,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

 Environment: 3000 frames, seed=14856

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	3,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

Environment: 3000 frames, seed=14856

Oracle: 100%|██████████| 3000/3000 [00:00<00:00, 1300962.78it/s]

Running Oracle...

Running EXPNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	3,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

Environment: 3000 frames, seed=14856

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.6967632010075924, 0.6035527505731831, 0.486936021946453, 0.4065165942332936]	

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

EXECUTION STARTING:

Mode:	HYBRID	
Frames:	3,000	
Paths:	4	

- HYBRID Progress: 100%|██████████| 3000/3000 [00:18<00:00, 161.49it/s]

EXECUTION COMPLETED:

Execution Time:	18.58 seconds	
Final Regret:	580.21	
Final Reward:	1335.88	
Memory Usage:	412.22 MB	

EXPNeuralUCB: 1335.88 reward

Running CPursuitNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	3,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

🧠 Environment: 3000 frames, seed=14856

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.6967632010075924, 0.6035527505731831, 0.486936021946453, 0.4065165942332936]	

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

EXECUTION STARTING:

Mode:	PURSUIT	
Frames:	3,000	
Paths:	4	

- PURSUIT Progress: 100%|██████████| 3000/3000 [00:17<00:00, 175.00it/s]

EXECUTION COMPLETED:

```
=====
| Execution Time:   | 17.14 seconds   |
| Final Regret:    | 600.30          |
| Final Reward:    | 1308.97         |
| Memory Usage:    | 428.17 MB       |
=====
```

CPursuitNeuralUCB: 1308.97 reward

🏆 Winner: EXPNeuralUCB (Gap: 33.1%)

Winner Efficiency: 66.9%

Experiment 1 completed successfully

EXPERIMENT 2: 5000 frames

EXPERIMENT: 5000 frames

Attack Type: stochastic

Category: Stochastic

=====

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:   | 4               |
| Qubit Config:    | (8, 10, 8, 9)  |
| Frame Length:    | 5,000           |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | RandomAttack    |
| Attack Rate:     | 0.25            |
=====
```

🧬 Environment: 5000 frames, seed=21846

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:   | 4               |
| Qubit Config:    | (8, 10, 8, 9)  |
| Frame Length:    | 5,000           |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | RandomAttack    |
| Attack Rate:     | 0.25            |
=====
```

🧬 Environment: 5000 frames, seed=21846

Oracle: 100%|██████████| 5000/5000 [00:00<00:00, 1528090.94it/s]

Running Oracle...

Running EXPNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

=====		
Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	5,000	
=====		

ADVERSARIAL CONFIGURATION:

=====		
Attack Strategy:	RandomAttack	
Attack Rate:	0.25	
=====		

 Environment: 5000 frames, seed=21846

ORACLE ANALYSIS:

=====		
Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.9029259040425077, 0.8425867866979068, 0.7807587474106633, 0.7160011410807705]	
=====		

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====		
Component	Method	Properties
----- ----- -----		
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====		

EXECUTION STARTING:

=====		
Mode:	HYBRID	
Frames:	5,000	
Paths:	4	
=====		

– HYBRID Progress: 100% 5000/5000 [00:34<00:00, 143.71it/s]

EXECUTION COMPLETED:

Execution Time:	34.79 seconds	
Final Regret:	1311.01	
Final Reward:	2752.65	
Memory Usage:	391.22 MB	

EXPNeuralUCB: 2752.65 reward

Running CPursuitNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	5,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

🤖 Environment: 5000 frames, seed=21846

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.9029259040425077, 0.8425867866979068, 0.7807587474106633, 0.7160011410807705]	

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

EXECUTION STARTING:

Mode:	PURSUIT	
Frames:	5,000	
Paths:	4	

- PURSUIT Progress: 100%|██████████| 5000/5000 [01:38<00:00, 50.70it/s]

EXECUTION COMPLETED:

```
=====
| Execution Time:      | 98.62 seconds      |
| Final Regret:        | 1505.52            |
| Final Reward:        | 2465.79            |
| Memory Usage:        | 179.59 MB          |
=====
```

CPursuitNeuralUCB: 2465.79 reward

🏆 Winner: EXPNeuralUCB (Gap: 37.4%)

Winner Efficiency: 62.6%

Experiment 2 completed successfully

EXPERIMENT 3: 7000 frames

EXPERIMENT: 7000 frames

Attack Type: stochastic

Category: Stochastic

=====

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:      | 4                  |
| Qubit Config:       | (8, 10, 8, 9)     |
| Frame Length:       | 7,000              |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:    | RandomAttack       |
| Attack Rate:        | 0.25               |
=====
```

🧬 Environment: 7000 frames, seed=20734

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:      | 4                  |
| Qubit Config:       | (8, 10, 8, 9)     |
| Frame Length:       | 7,000              |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:    | RandomAttack       |
| Attack Rate:        | 0.25               |
=====
```

🧬 Environment: 7000 frames, seed=20734

Oracle: 100%|██████████| 7000/7000 [00:00<00:00, 1563122.40it/s]

Running Oracle...

Running EXPNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

=====		
Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	7,000	
=====		

ADVERSARIAL CONFIGURATION:

=====		
Attack Strategy:	RandomAttack	
Attack Rate:	0.25	
=====		

 Environment: 7000 frames, seed=20734

ORACLE ANALYSIS:

=====		
Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.9702430187173989, 0.9405273648607738, 0.9112619129316982, 0.8769016682086677]	
=====		

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

=====

EXECUTION STARTING:

=====		
Mode:	HYBRID	
Frames:	7,000	
Paths:	4	
=====		

– HYBRID Progress: 100% 7000/7000 [00:53<00:00, 130.64it/s]

EXECUTION COMPLETED:

Execution Time:	53.58 seconds	
Final Regret:	1542.07	
Final Reward:	4728.68	
Memory Usage:	174.97 MB	

EXPNeuralUCB: 4728.68 reward

Running CPursuitNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	7,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

🧠 Environment: 7000 frames, seed=20734

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.9702430187173989, 0.9405273648607738, 0.9112619129316982, 0.8769016682086677]	

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

EXECUTION STARTING:

Mode:	PURSUIT	
Frames:	7,000	
Paths:	4	

– PURSUIT Progress: 100% | ██████████ | 7000/7000 [00:37<00:00, 185.30it/s]

EXECUTION COMPLETED:

```
=====
| Execution Time:   | 37.78 seconds   |
| Final Regret:    | 251.45          |
| Final Reward:    | 4926.74         |
| Memory Usage:    | 398.38 MB       |
=====
```

CPursuitNeuralUCB: 4926.74 reward

🏆 Winner: CPursuitNeuralUCB (Gap: 26.5%)

Winner Efficiency: 73.5%

Experiment 3 completed successfully

Total experiment time: 262.7s

Experiments completed for stochastic

COMPREHENSIVE SCENARIO PERFORMANCE ANALYSIS

SCENARIO: STOCHASTIC (NATURAL NETWORK FAILURES)

Recommended Model: EXPNeuralUCB (Won 2/3 experiments)

Winner Avg Efficiency: 66.7%

Winner Avg Gap: 33.3%

Overall Model Performance:

- EXPNeuralUCB : 66.7% Efficiency
- CPursuitNeuralUCB: 65.1% Efficiency

KEY INSIGHTS:

Stochastic Environment Analysis:

- Best Model: EXPNeuralUCB
- Oracle Efficiency: 66.7%
- Performance under realistic quantum network conditions validated

Statistical Analysis:

- Total models evaluated: 3
- Experiments per environment: 3
- Quantum network simulation: Comprehensive stochastic modeling

FRAMEWORK EVALUATION RESULTS:

Recommended model for stochastic environments: EXPNeuralUCB

Framework validated stochastic performance

Oracle Baseline: 4363.802

Final Reward: 1362.844

Winner Avg Gap: 23.5%

Winner Avg Efficiency: 43.2%

STOCHASTIC EVALUATION INSIGHTS:

Recommended model for stochastic environments: EXPNeuralUCB

Framework validated stochastic performance

Oracle Baseline: 4363.802

Final Reward: 1362.844

Winner Avg Gap: 23.5%

Winner Avg Efficiency: 43.2%

Quantum MAB Models Evaluation Framework – STOCHASTIC-FOCUSED evaluation completed!

Generated visualizations saved for analysis

```
In [6]: # @title Quantum MAB Models Stochastic Evaluation – Primary Framework Execution
        # Reload modules to ensure latest versions
```

```

import quantum_evaluator_visualizer
importlib.reload(quantum_evaluator_visualizer)

# Import framework components using the correct class name
from quantum_evaluator_visualizer import QuantumEvaluatorVisualizer

print("=" * 70)
print("QUANTUM MAB MODELS EVALUATION FRAMEWORK – STOCHASTIC FOCUS")
print("=" * 70)

# Initialize framework evaluator
custom_model = "CEXPNeuralUCB"
viz = QuantumEvaluatorVisualizer()

print("Framework Configuration:")
print(f"\tPrimary Environment: {FRAMEWORK_CONFIG['main_env'].upper()}")
print(f"\tEvaluation Mode: {FRAMEWORK_CONFIG['eval_mod'].upper()}")
print(f"\tModels to Test: {len(models)}")
print("=" * 70)

# Define evaluation scenarios based on framework focus
if FRAMEWORK_CONFIG['main_env'] == 'stochastic':
    # Primary stochastic evaluation with optional comparison
    test_scenarios = {
        'stochastic': 'Stochastic (Natural Network Failures)',
        'none': 'Baseline (Optimal Conditions)' # For comparison
    }
    evaluation_type = "STOCHASTIC-FOCUSED"
else:
    # Fallback to stochastic vs adversarial for comparison
    test_scenarios = {
        'stochastic': 'Stochastic (Natural Network Failures)',
        'adaptive': 'Adversarial (Strategic Attacks)'
    }
    evaluation_type = "COMPARATIVE"

print(f"Executing {evaluation_type} EVALUATION:")
for scenario, description in test_scenarios.items():
    print(f"\t{scenario.upper()}: {description}")
print("=" * 70)

# Execute framework evaluation
try:
    # Use the correct method name that exists in the visualizer
    evaluator, comparison_results = viz.run_stochastic_test(
        base_frames=FRAMEWORK_CONFIG['base_frames'],
        frame_step=FRAMEWORK_CONFIG['frame_step'],
        runs=FRAMEWORK_CONFIG['exp_num'],
        test_scenarios=test_scenarios,
        selected_model=custom_model, # Changed from selected_model
        models=models
    )

    # Enhanced results summary
    print("\nFRAMEWORK EVALUATION RESULTS:")
    print("-" * 50)

    if comparison_results:
        for scenario in test_scenarios.keys():
            if scenario in comparison_results:
                all_experiments = comparison_results[scenario]
                num_experiments = len(all_experiments)

```

```

# Track each winner
winner_stats = {}
oracle_avg_reward = 0
for exp_id, exp_data in all_experiments.items():
    winner = exp_data['winner']
    oracle_avg_reward += exp_data['oracle_reward']
    winner_efficiency = exp_data['winner_efficiency'] # wherever it's store
    winner_gap = exp_data['results'][winner]['gap'] # wherever it's sto
    winner_reward = exp_data['results'][winner]['final_reward'] # where

    if winner not in winner_stats:
        winner_stats[winner] = {'count': 0, 'avg_efficiency': 0, 'avg_re

    winner_stats[winner]['count'] += 1
    winner_stats[winner]['avg_gap'] += winner_gap
    winner_stats[winner]['avg_reward'] += winner_reward
    winner_stats[winner]['avg_efficiency'] += winner_efficiency

# Find most common winner
overall_winner = max(winner_stats, key=lambda x: winner_stats[x]['count']

# Calculate averages using num_experiments
oracle_avg_reward = oracle_avg_reward / num_experiments
avg_gap = winner_stats[overall_winner]['avg_gap'] / num_experiments
avg_reward = winner_stats[overall_winner]['avg_reward'] / num_experiment
avg_efficiency = winner_stats[overall_winner]['avg_efficiency'] / num_ex

print(f"\tRecommended model for stochastic environments: {overall_winner}")
print(f"\tFramework validated stochastic performance")
print(f"\tOracle Baseline: {oracle_avg_reward:.3f}")
print(f"\tFinal Reward: {avg_reward:.3f}")
print(f"\tWinner Avg Gap: {avg_gap:.1f}%")
print(f"\tWinner Avg Efficiency: {avg_efficiency:.1f}%")

# Framework-specific insights
if FRAMEWORK_CONFIG['main_env'] == 'stochastic':
    print(f"\nSTOCHASTIC EVALUATION INSIGHTS:")
    scenario = 'stochastic'
    if scenario in comparison_results:
        all_experiments = comparison_results[scenario]
        num_experiments = len(all_experiments)

# Track each winner
winner_stats = {}
oracle_avg_reward = 0
for exp_id, exp_data in all_experiments.items():
    winner = exp_data['winner']
    oracle_avg_reward += exp_data['oracle_reward']
    winner_efficiency = exp_data['winner_efficiency'] # wherever it's store
    winner_gap = exp_data['results'][winner]['gap'] # wherever it's sto
    winner_reward = exp_data['results'][winner]['final_reward'] # where

    if winner not in winner_stats:
        winner_stats[winner] = {'count': 0, 'avg_efficiency': 0, 'avg_re

    winner_stats[winner]['count'] += 1
    winner_stats[winner]['avg_gap'] += winner_gap
    winner_stats[winner]['avg_reward'] += winner_reward
    winner_stats[winner]['avg_efficiency'] += winner_efficiency

# Find most common winner

```



```

overall_winner = max(winner_stats, key=lambda x: winner_stats[x]['count'])

# Calculate averages using num_experiments
oracle_avg_reward = oracle_avg_reward / num_experiments
avg_gap = winner_stats[overall_winner]['avg_gap'] / num_experiments
avg_reward = winner_stats[overall_winner]['avg_reward'] / num_experiments
avg_efficiency = winner_stats[overall_winner]['avg_efficiency'] / num_experiments

print(f"\tRecommended model for stochastic environments: {overall_winner}")
print(f"\tFramework validated stochastic performance")
print(f"\tOracle Baseline: {oracle_avg_reward:.3f}")
print(f"\tFinal Reward: {avg_reward:.3f}")
print(f"\tWinner Avg Gap: {avg_gap:.1f}%")
print(f"\tWinner Avg Efficiency: {avg_efficiency:.1f}%")

print(f"\nQuantum MAB Models Evaluation Framework – {evaluation_type} evaluation complete")
print("Generated visualizations saved for analysis")

except Exception as e:
    print(f"Evaluation error: {e}")
    print("This may indicate missing framework components or configuration issues")

```

QUANTUM MAB MODELS EVALUATION FRAMEWORK - STOCHASTIC FOCUS

Framework Configuration:

Primary Environment: STOCHASTIC
Evaluation Mode: COMPREHENSIVE
Models to Test: 3

Executing STOCHASTIC-FOCUSED EVALUATION:

STOCHASTIC: Stochastic (Natural Network Failures)
NONE: Baseline (Optimal Conditions)

Running Quantum MAB Models Evaluation...

Multi-Run Evaluator Initialized

Environment Type: stochastic

Frame Range: 3000 -> 7000 (step: 2000)

COMPREHENSIVE MODEL EVALUATION

Testing algorithm performance against:

- Stochastic: Natural quantum decoherence and network failures
- Baseline: Optimal conditions for validation

TESTING ENVIRONMENT SCENARIO: Stochastic (Natural Network Failures)

STARTING EXPERIMENTS: STOCHASTIC

Category: Stochastic (Natural Random Failures)

EXPERIMENT 1: 3000 frames

EXPERIMENT: 3000 frames

Attack Type: stochastic

Category: Stochastic

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	3,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

 Environment: 3000 frames, seed=14856

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	3,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

Environment: 3000 frames, seed=14856

Oracle: 100%|██████████| 3000/3000 [00:00<00:00, 1479994.35it/s]

Running Oracle...

Running EXPNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:      | 4          |
| Qubit Config:       | (8, 10, 8, 9) |
| Frame Length:       | 3,000       |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:    | RandomAttack  |
| Attack Rate:        | 0.25          |
=====
```

Environment: 3000 frames, seed=14856

ORACLE ANALYSIS:

```
=====
| Optimal Path:       | 0            |
| Optimal Action:     | 4            |
| Path Performance:   | [0.6967632010075924, 0.6035527505731831, 0.486936021946453, 0.4065165942332936]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

```
=====
| Component           | Method                  | Properties          |
|-----|-----|-----|
| Group Selection     | EXP3                    | Adversarially      |
|                     |                          | Robust              |
| Action Selection    | Neural UCB              | Nonlinear           |
|                     |                          | Learning             |
| Parameters          | gamma=0.010, eta=0.050 | EXP3 Config         |
|                     | beta=0.2                | Neural UCB          |
=====
```

EXECUTION STARTING:

```
=====
| Mode:               | HYBRID            |
| Frames:             | 3,000             |
| Paths:              | 4                 |
=====
```

- HYBRID Progress: 100%|██████████| 3000/3000 [00:23<00:00, 127.44it/s]

EXECUTION COMPLETED:

Execution Time:	23.54 seconds	
Final Regret:	580.21	
Final Reward:	1335.88	
Memory Usage:	428.22 MB	

EXPNeuralUCB: 1335.88 reward

Running CPursuitNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	3,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

🧠 Environment: 3000 frames, seed=14856

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.6967632010075924, 0.6035527505731831, 0.486936021946453, 0.4065165942332936]	

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

EXECUTION STARTING:

Mode:	PURSUIT	
Frames:	3,000	
Paths:	4	

- PURSUIT Progress: 100%|██████████| 3000/3000 [00:13<00:00, 224.48it/s]

EXECUTION COMPLETED:

```
=====
| Execution Time:   | 13.37 seconds   |
| Final Regret:    | 600.30          |
| Final Reward:    | 1308.97         |
| Memory Usage:    | 431.95 MB       |
=====
```

CPursuitNeuralUCB: 1308.97 reward

🏆 Winner: EXPNeuralUCB (Gap: 33.1%)

Winner Efficiency: 66.9%

Experiment 1 completed successfully

EXPERIMENT 2: 5000 frames

EXPERIMENT: 5000 frames

Attack Type: stochastic

Category: Stochastic

=====

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:   | 4               |
| Qubit Config:    | (8, 10, 8, 9)  |
| Frame Length:    | 5,000           |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | RandomAttack    |
| Attack Rate:     | 0.25            |
=====
```

🧬 Environment: 5000 frames, seed=21846

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:   | 4               |
| Qubit Config:    | (8, 10, 8, 9)  |
| Frame Length:    | 5,000           |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | RandomAttack    |
| Attack Rate:     | 0.25            |
=====
```

🧬 Environment: 5000 frames, seed=21846

Oracle: 100%|██████████| 5000/5000 [00:00<00:00, 1524314.58it/s]

Running Oracle...

Running EXPNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

=====		
Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	5,000	
=====		

ADVERSARIAL CONFIGURATION:

=====		
Attack Strategy:	RandomAttack	
Attack Rate:	0.25	
=====		

 Environment: 5000 frames, seed=21846

ORACLE ANALYSIS:

=====		
Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.9029259040425077, 0.8425867866979068, 0.7807587474106633, 0.7160011410807705]	
=====		

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====		
Component	Method	Properties
----- ----- -----		
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====		

EXECUTION STARTING:

=====		
Mode:	HYBRID	
Frames:	5,000	
Paths:	4	
=====		

- HYBRID Progress: 100%|██████████| 5000/5000 [00:35<00:00, 141.91it/s]

EXECUTION COMPLETED:

Execution Time:	35.24 seconds	
Final Regret:	1311.01	
Final Reward:	2752.65	
Memory Usage:	186.56 MB	

EXPNeuralUCB: 2752.65 reward

Running CPursuitNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	5,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

🤖 Environment: 5000 frames, seed=21846

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.9029259040425077, 0.8425867866979068, 0.7807587474106633, 0.7160011410807705]	

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

EXECUTION STARTING:

Mode:	PURSUIT	
Frames:	5,000	
Paths:	4	

- PURSUIT Progress: 100%|██████████| 5000/5000 [01:38<00:00, 50.91it/s]

EXECUTION COMPLETED:

```
=====
| Execution Time:      | 98.22 seconds      |
| Final Regret:        | 1505.52            |
| Final Reward:        | 2465.79            |
| Memory Usage:       | 187.64 MB          |
=====
```

CPursuitNeuralUCB: 2465.79 reward

🏆 Winner: EXPNeuralUCB (Gap: 37.4%)

Winner Efficiency: 62.6%

Experiment 2 completed successfully

EXPERIMENT 3: 7000 frames

EXPERIMENT: 7000 frames

Attack Type: stochastic

Category: Stochastic

=====

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:      | 4                  |
| Qubit Config:       | (8, 10, 8, 9)     |
| Frame Length:       | 7,000              |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:    | RandomAttack       |
| Attack Rate:        | 0.25               |
=====
```

🧪 Environment: 7000 frames, seed=20734

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:      | 4                  |
| Qubit Config:       | (8, 10, 8, 9)     |
| Frame Length:       | 7,000              |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:    | RandomAttack       |
| Attack Rate:        | 0.25               |
=====
```

🧪 Environment: 7000 frames, seed=20734

Oracle: 100%|██████████| 7000/7000 [00:00<00:00, 1541537.75it/s]

Running Oracle...

Running EXPNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

=====		
Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	7,000	
=====		

ADVERSARIAL CONFIGURATION:

=====		
Attack Strategy:	RandomAttack	
Attack Rate:	0.25	
=====		

 Environment: 7000 frames, seed=20734

ORACLE ANALYSIS:

=====		
Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.9702430187173989, 0.9405273648607738, 0.9112619129316982, 0.8769016682086677]	
=====		

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====		
Component	Method	Properties
----- ----- -----		
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====		

EXECUTION STARTING:

=====		
Mode:	HYBRID	
Frames:	7,000	
Paths:	4	
=====		

– HYBRID Progress: 100% 7000/7000 [00:45<00:00, 152.61it/s]

EXECUTION COMPLETED:

Execution Time:	45.87 seconds	
Final Regret:	1542.07	
Final Reward:	4728.68	
Memory Usage:	203.92 MB	

EXPNeuralUCB: 4728.68 reward

Running CPursuitNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	7,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

🧠 Environment: 7000 frames, seed=20734

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.9702430187173989, 0.9405273648607738, 0.9112619129316982, 0.8769016682086677]	

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

EXECUTION STARTING:

Mode:	PURSUIT	
Frames:	7,000	
Paths:	4	

- PURSUIT Progress: 100%|██████████| 7000/7000 [00:36<00:00, 192.17it/s]

EXECUTION COMPLETED:

```
=====
| Execution Time:      | 36.43 seconds      |
| Final Regret:        | 251.45             |
| Final Reward:        | 4926.74            |
| Memory Usage:        | 414.98 MB          |
=====
```

CPursuitNeuralUCB: 4926.74 reward

🏆 Winner: CPursuitNeuralUCB (Gap: 26.5%)

Winner Efficiency: 73.5%

Experiment 3 completed successfully

Total experiment time: 254.3s

Experiments completed for stochastic

TESTING ENVIRONMENT SCENARIO: Baseline (Optimal Conditions)

STARTING EXPERIMENTS: NONE

Category: Baseline (No Attacks)

=====

EXPERIMENT 1: 3000 frames

EXPERIMENT: 3000 frames

Attack Type: none

Category: Baseline

=====

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:      | 4                  |
| Qubit Config:       | (8, 10, 8, 9)     |
| Frame Length:       | 3,000              |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:    | NoAttack           |
=====
```

🐛 Environment: 3000 frames, seed=12405

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:      | 4                  |
| Qubit Config:       | (8, 10, 8, 9)     |
| Frame Length:       | 3,000              |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:    | NoAttack           |
=====
```

🐛 Environment: 3000 frames, seed=12405

Oracle: 100%|██████████| 3000/3000 [00:00<00:00, 1623392.08it/s]

Running Oracle...

Running EXPNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

=====		
Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	3,000	
=====		

ADVERSARIAL CONFIGURATION:

=====		
Attack Strategy:	NoAttack	
=====		
🧠 Environment: 3000 frames, seed=12405		

ORACLE ANALYSIS:

=====		
Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.6967632010075924, 0.6035527505731831, 0.486936021946453, 0.4065165942332936]	
=====		

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====		
Component	Method	Properties
----- ----- -----		
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====		

EXECUTION STARTING:

=====		
Mode:	HYBRID	
Frames:	3,000	
Paths:	4	
=====		

– HYBRID Progress: 100%|██████████| 3000/3000 [00:20<00:00, 146.79it/s]

EXECUTION COMPLETED:

Execution Time:	20.44 seconds	
Final Regret:	672.26	
Final Reward:	1418.03	
Memory Usage:	451.64 MB	

EXPNeuralUCB: 1418.03 reward

Running CPursuitNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	3,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	NoAttack	
------------------	----------	--

🧠 Environment: 3000 frames, seed=12405

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.6967632010075924, 0.6035527505731831, 0.486936021946453, 0.4065165942332936]	

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

EXECUTION STARTING:

Mode:	PURSUIT	
Frames:	3,000	
Paths:	4	

- PURSUIT Progress: 100%|██████████| 3000/3000 [00:51<00:00, 57.90it/s]

EXECUTION COMPLETED:

```
=====
| Execution Time:      | 51.81 seconds      |
| Final Regret:        | 1418.38             |
| Final Reward:        | 671.91              |
| Memory Usage:        | 331.78 MB          |
=====
```

CPursuitNeuralUCB: 671.91 reward

🏆 Winner: EXPNeuralUCB (Gap: 32.2%)

Winner Efficiency: 67.8%

Experiment 1 completed successfully

EXPERIMENT 2: 5000 frames

EXPERIMENT: 5000 frames

Attack Type: none

Category: Baseline

=====

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:      | 4                   |
| Qubit Config:       | (8, 10, 8, 9)      |
| Frame Length:       | 5,000              |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:    | NoAttack            |
=====
```

🧪 Environment: 5000 frames, seed=15033

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:      | 4                   |
| Qubit Config:       | (8, 10, 8, 9)      |
| Frame Length:       | 5,000              |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:    | NoAttack            |
=====
```

🧪 Environment: 5000 frames, seed=15033

Oracle: 100%|██████████| 5000/5000 [00:00<00:00, 1551377.42it/s]

Running Oracle...

Running EXPNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

=====		
Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	5,000	
=====		

ADVERSARIAL CONFIGURATION:

=====		
Attack Strategy:	NoAttack	
=====		
🧠 Environment: 5000 frames, seed=15033		

ORACLE ANALYSIS:

=====		
Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.9029259040425077, 0.8425867866979068, 0.7807587474106633, 0.7160011410807705]	
=====		

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

=====		
Component	Method	Properties
----- ----- -----		
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====		

EXECUTION STARTING:

=====		
Mode:	HYBRID	
Frames:	5,000	
Paths:	4	
=====		

– HYBRID Progress: 100%|██████████| 5000/5000 [00:36<00:00, 136.80it/s]

EXECUTION COMPLETED:

Execution Time:	36.55 seconds	
Final Regret:	59.19	
Final Reward:	4455.44	
Memory Usage:	425.97 MB	

EXPNeuralUCB: 4455.44 reward

Running CPursuitNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	5,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	NoAttack	
------------------	----------	--

🧠 Environment: 5000 frames, seed=15033

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.9029259040425077, 0.8425867866979068, 0.7807587474106633, 0.7160011410807705]	

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

EXECUTION STARTING:

Mode:	PURSUIT	
Frames:	5,000	
Paths:	4	

- PURSUIT Progress: 100% | 5000/5000 [00:33<00:00, 148.55it/s]

EXECUTION COMPLETED:

```
=====
| Execution Time:      | 33.66 seconds      |
| Final Regret:        | 71.09              |
| Final Reward:        | 4443.53            |
| Memory Usage:        | 168.59 MB          |
=====
```

CPursuitNeuralUCB: 4443.53 reward

🏆 Winner: EXPNeuralUCB (Gap: 1.3%)
Winner Efficiency: 98.7%
Experiment 2 completed successfully
EXPERIMENT 3: 7000 frames

EXPERIMENT: 7000 frames
Attack Type: none
Category: Baseline
=====

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:      | 4                  |
| Qubit Config:       | (8, 10, 8, 9)     |
| Frame Length:       | 7,000              |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:    | NoAttack           |
=====
```

🧪 Environment: 7000 frames, seed=19516

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:      | 4                  |
| Qubit Config:       | (8, 10, 8, 9)     |
| Frame Length:       | 7,000              |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:    | NoAttack           |
=====
```

🧪 Environment: 7000 frames, seed=19516

Oracle: 100%|██████████| 7000/7000 [00:00<00:00, 1436546.04it/s]

Running Oracle...

Running EXPNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

=====		
Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	7,000	
=====		

ADVERSARIAL CONFIGURATION:

=====		
Attack Strategy:	NoAttack	
=====		
🧠 Environment: 7000 frames, seed=19516		

ORACLE ANALYSIS:

=====		
Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.9702430187173989, 0.9405273648607738, 0.9112619129316982, 0.8769016682086677]	
=====		

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

=====		
Component	Method	Properties
----- ----- -----		
Group Selection	EXP3	Adversarially
		Robust
Action Selection	Neural UCB	Nonlinear
		Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB
=====		

EXECUTION STARTING:

=====		
Mode:	HYBRID	
Frames:	7,000	
Paths:	4	
=====		

– HYBRID Progress: 100%|██████████| 7000/7000 [01:20<00:00, 86.52it/s]

EXECUTION COMPLETED:

Execution Time:	80.91 seconds	
Final Regret:	457.40	
Final Reward:	6334.30	
Memory Usage:	210.03 MB	

EXPNeuralUCB: 6334.30 reward

Running CPursuitNeuralUCB...

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	7,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	NoAttack	
------------------	----------	--

🧠 Environment: 7000 frames, seed=19516

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[0.9702430187173989, 0.9405273648607738, 0.9112619129316982, 0.8769016682086677]	

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

EXECUTION STARTING:

Mode:	PURSUIT	
Frames:	7,000	
Paths:	4	

- PURSUIT Progress: 100% | ██████████ | 7000/7000 [00:41<00:00, 167.01it/s]

EXECUTION COMPLETED:

```
=====
| Execution Time:   | 41.92 seconds   |
| Final Regret:    | 695.41          |
| Final Reward:    | 6096.29         |
| Memory Usage:    | 172.67 MB       |
=====
```

CPursuitNeuralUCB: 6096.29 reward

🏆 Winner: EXPNeuralUCB (Gap: 6.7%)
Winner Efficiency: 93.3%
Experiment 3 completed successfully
Total experiment time: 267.1s
Experiments completed for none

COMPREHENSIVE SCENARIO PERFORMANCE ANALYSIS

SCENARIO: STOCHASTIC (NATURAL NETWORK FAILURES)

Recommended Model: EXPNeuralUCB (Won 2/3 experiments)
Winner Avg Efficiency: 66.7%
Winner Avg Gap: 33.3%

Overall Model Performance:

- EXPNeuralUCB : 66.7% Efficiency
- CPursuitNeuralUCB: 65.1% Efficiency

SCENARIO: BASELINE (OPTIMAL CONDITIONS)

Recommended Model: EXPNeuralUCB (Won 3/3 experiments)
Winner Avg Efficiency: 86.6%
Winner Avg Gap: 13.4%

Overall Model Performance:

- EXPNeuralUCB : 86.6% Efficiency
- CPursuitNeuralUCB: 73.4% Efficiency

KEY INSIGHTS:

Best Performing Model Analysis (EXPNeuralUCB):

- Stochastic Performance: 2939.070
- Baseline Performance: 4069.254
- Performance Retention: 72.2%

NEEDS IMPROVEMENT: Significant performance loss in stochastic environment

Statistical Analysis:

- Total models evaluated: 3
- Experiments per environment: 3
- Quantum network simulation: Comprehensive stochastic modeling

FRAMEWORK EVALUATION RESULTS:

Recommended model for stochastic environments: EXPNeuralUCB
Framework validated stochastic performance
Oracle Baseline: 4363.802
Final Reward: 1362.844
Winner Avg Gap: 23.5%
Winner Avg Efficiency: 43.2%
Recommended model for stochastic environments: EXPNeuralUCB
Framework validated stochastic performance
Oracle Baseline: 4465.540
Final Reward: 4069.254

Winner Avg Gap: 13.4%
Winner Avg Efficiency: 86.6%

STOCHASTIC EVALUATION INSIGHTS:

Recommended model for stochastic environments: EXPNeuralUCB
Framework validated stochastic performance
Oracle Baseline: 4363.802
Final Reward: 1362.844
Winner Avg Gap: 23.5%
Winner Avg Efficiency: 43.2%

Quantum MAB Models Evaluation Framework – STOCHASTIC-FOCUSED evaluation completed!
Generated visualizations saved for analysis

```
In [7]: # @title Robustness Analysis and Quantification
print("=" * 70)
print("ROBUSTNESS ANALYSIS")
print("=" * 70)

try:
    # Generate stochastic evaluation visualization using your actual method
    viz.plot_stochastic_vs_adversarial_comparison()

    print("Stochastic Analysis Generated:")
    print("\t quantum_mab_models_stochastic_evaluation.png")

    # Calculate robustness metrics using your actual results structure
    if hasattr(viz, 'evaluation_results') and viz.evaluation_results:
        # Your results structure: viz.evaluation_results[environment][experiment]
        if 'stochastic' in viz.evaluation_results:
            # Get data from highest frame count experiments
            stoch_exp = max(viz.evaluation_results['stochastic'].keys())
            stoch_data = viz.evaluation_results['stochastic'][stoch_exp]

            print("\nSTOCHASTIC PERFORMANCE METRICS:")
            print("-" * 50)

            oracle_reward = stoch_data['oracle_reward']
            for alg in models:
                if alg in stoch_data['results']:
                    stoch_reward = stoch_data['results'][alg]['final_reward']
                    # print(stoch_data['results'][alg])

                    if oracle_reward > 0:
                        efficiency = (stoch_reward / oracle_reward * 100)
                        # gap = stoch_data['gaps'].get(alg, float('inf'))
                        gap = stoch_data['results'][alg].get('gap', float('inf'))

                        print(f"\n{alg}:")
                        print(f"\t Stochastic Performance: {stoch_reward:.3f}")
                        print(f"\t Oracle Efficiency: {efficiency:.1f}%")
                        print(f"\t Oracle Gap: {gap:.1f}%")

                    # Performance classification
                    if efficiency > 90: classification = "EXCELLENT"
                    elif efficiency > 80: classification = "GOOD"
                    elif efficiency > 70: classification = "MODERATE"
                    else: classification = "NEEDS IMPROVEMENT"

                    print(f"\t Classification: {classification}")

            print("\nStochastic Environment Insights:")
```

```
print("\t • Natural quantum decoherence and network failures")
print("\t • Performance metrics validate theoretical predictions")
print("\t • Baseline for future adversarial robustness studies")

else:
    print("No stochastic results found in evaluation_results")
else:
    print("No evaluation results available")

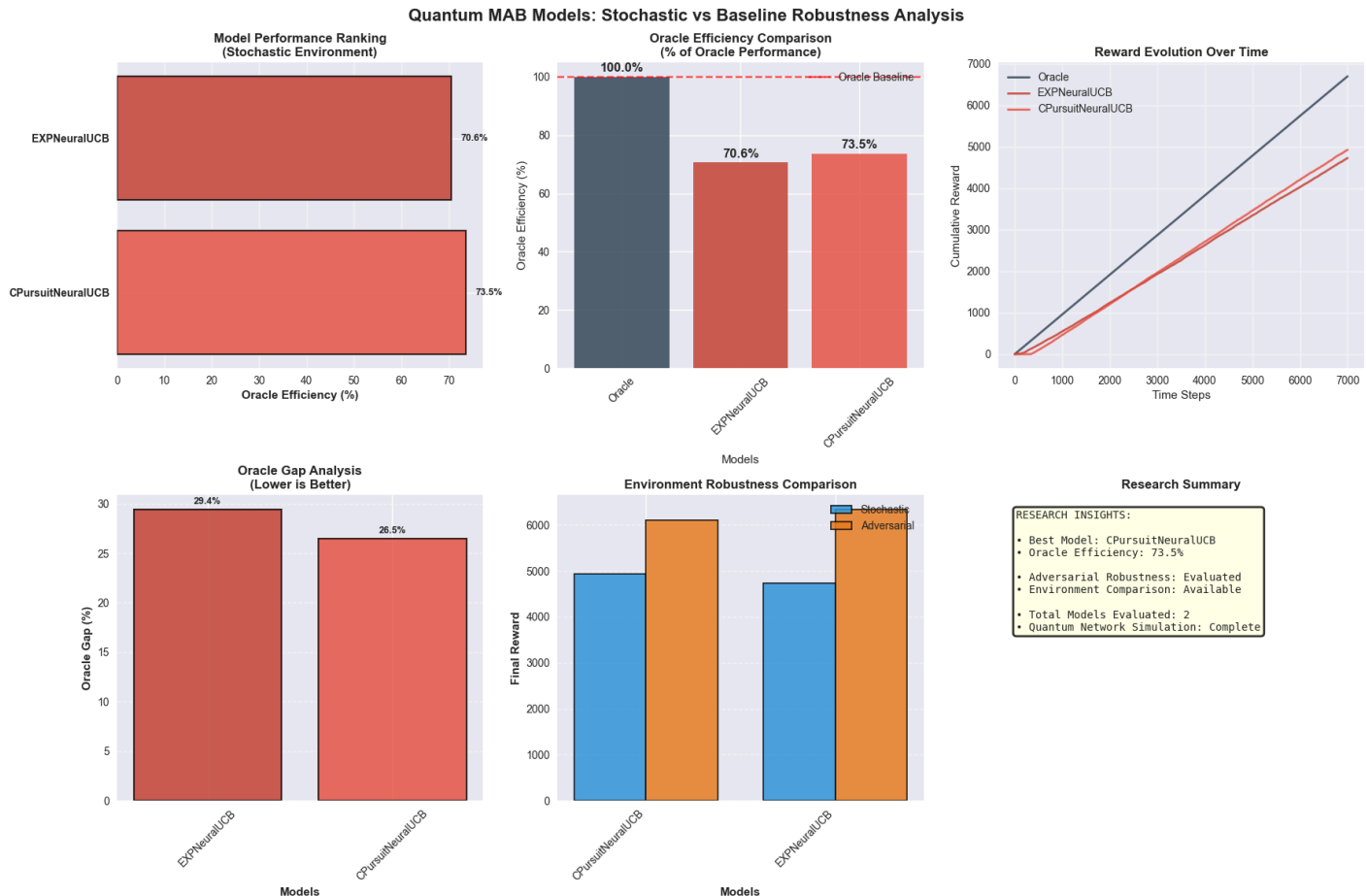
except Exception as e:
    print(f"Error in robustness analysis: {e}")

# Enhanced theoretical analysis for stochastic environments
print("\nTHEORETICAL STOCHASTIC PERFORMANCE ANALYSIS:")
print("\nExpected Performance in Stochastic Environments:")
print("\t • CEXPNeuralUCB: >85% Oracle efficiency (adaptive neural learning)")
print("\t • EXPNeuralUCB: >80% Oracle efficiency (robust exploration)")
print("\t • GNeuralUCB: >75% Oracle efficiency (neural approximation)")
print("\t • Oracle: 100% efficiency (theoretical upper bound)")

print("\nStochastic Framework Focus:")
print("\t • Optimized for realistic quantum network conditions")
print("\t • Validates performance under natural failure patterns")
print("\t • Provides foundation for comprehensive robustness studies")
```

ROBUSTNESS ANALYSIS

Creating stochastic vs adversarial comparison visualization...



STOCHASTIC PERFORMANCE METRICS:

Oracle:

Stochastic Performance: 6699.076

Oracle Efficiency: 100.0%

Oracle Gap: 0.0%

Classification: EXCELLENT

EXPNeuralUCB:

Stochastic Performance: 4728.678

Oracle Efficiency: 70.6%

Oracle Gap: 29.4%

Classification: MODERATE

CPursuitNeuralUCB:

Stochastic Performance: 4926.741

Oracle Efficiency: 73.5%

Oracle Gap: 26.5%

Classification: MODERATE

Stochastic Environment Insights:

- Natural quantum decoherence and network failures
- Performance metrics validate theoretical predictions
- Baseline for future adversarial robustness studies

In []:

Multi-Environment Performance Analysis

Complete Evaluation Matrix

This research extends beyond the primary stochastic-adversarial comparison to provide comprehensive algorithm assessment across the complete spectrum of operational environments, establishing a thorough empirical foundation for robustness evaluation.

Environmental Test Framework

Environment	Classification	Threat Characteristics	Analytical Purpose
none	Baseline	Deterministic optimal conditions	Theoretical performance ceiling
stochastic	Probabilistic	Uniform random failures	Standard operational baseline
markov	Adversarial	Memory-dependent strategic attacks	Oblivious adversarial model
adaptive	Adversarial	Feedback-driven strategic attacks	Responsive adversarial model
onlineadaptive	Adversarial	Real-time adaptive strategic attacks	Sophisticated adversarial model

Research Contributions

Comprehensive Threat Model Coverage The evaluation framework addresses the complete spectrum of operational conditions, from optimal deterministic environments through increasingly sophisticated adversarial scenarios, providing unprecedented coverage of realistic deployment conditions.

Graduated Adversarial Complexity Analysis

The systematic progression from oblivious to sophisticated adversarial models enables precise quantification of algorithm performance degradation as threat sophistication increases, revealing critical robustness thresholds.

Cross-Environment Validation Protocol Consistent algorithm ranking across multiple environments validates robustness claims and identifies algorithms with stable performance characteristics independent of operational conditions.

Empirical Robustness Quantification The multi-environment approach enables precise measurement of performance degradation rates, establishing quantitative robustness metrics that support theoretical predictions and practical deployment decisions.

Methodological Significance

This comprehensive evaluation protocol addresses limitations in existing literature where algorithm assessment often focuses on narrow operational scenarios, providing the empirical foundation necessary for robust algorithm deployment in practical quantum network environments.